

CAB302 – FarmBook

Idontknow

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I. OVERVIEW

Project name: FarmBook

Project type: Farm inventory and farm-operations management desktop application.

Target users: Farmers / farm operators who need one place to monitor crops, stock, livestock and selected farm resources.

Problem statement: FarmBook addresses the difficulty of keeping farm information organised across crops, inventory, livestock, resource use and maintenance. The project aims to place these records in one system so a farmer can check the current state of the farm more easily and make more informed operational decisions.

Proposed solution: FarmBook will provide a front end that farmers can use to view and update farm information, supported by a back end that stores and manages operational records such as stock movements, livestock data, crop progress, water/energy use and waste records.

Technology for Sustainable Futures: The strongest sustainability connection in the current idea is resource efficiency. By helping farmers track inventory, waste, water and energy use, FarmBook can support better visibility of resource consumption and more informed farm-management decisions. The project can also support economic sustainability by improving stock control and reducing unnecessary resource use.

II. PROJECT SCOPE

A. Core Features

1. Authentication

- Account access
- Login
- Logout

2. Crop Management

- Record planted crops/plants.

- Record crop growth/progress.
- Record crop needs or care information.

3. *Inventory & Stock Management*

- Manage stock of seeds and equipment.
- Record stock-in and stock-out movements.
- Provide a user-facing view of current stock and progress.

4. *Livestock Management*

- Record and manage animals.
- Record animal feed information.
- Record useful farm outputs such as cow milk, sheep wool and chicken eggs.

5. *Sustainability & Farm Resource Tracking*

- Record waste-management information.
- Record energy-consumption rates.
- Record water-consumption rates and relevant offsets where appropriate.
- Provide simple summaries so users can check farm resource use.

B. Future / Optional Features

1. *Maintenance*

- Record maintenance needs
- Record completed maintenance activities

2. *Networking and Security*

- Potential future support for resource or stock coordination between farms.
- Potential future sensor-related alerts for farm security or predator detection.

III. FUNCTIONAL REQUIREMENTS

Core Feature	Functional Requirements (FR)	Description
Authentication	FR01 – User Access	The system shall allow registered users to securely access FarmBook.
	FR02 – Crop Records	The system shall allow users to record crops planted on the farm.

Crop Management	FR03 – Crop Progress	The system shall allow users to record and update crop growth or progress information.
	FR04 – Crop Needs	The system shall allow users to record relevant crop-care information.
Inventory	FR05 – Inventory Records	The system shall allow users to maintain records of seeds and farm equipment.
	FR06 – Stock Movement	The system shall allow users to record stock-in and stock-out movements.
Livestock	FR07 – Livestock Records	The system shall allow users to create and maintain livestock records.
	FR08 – Feed Information	The system shall allow users to record relevant animal-feed information.
	FR09 – Livestock Outputs	The system shall allow users to record farm outputs such as milk, wool and eggs where applicable.
Sustainability	FR10 – Resource Consumption	The system shall allow users to record water and energy-consumption information.
	FR11 – Waste Records	The system shall allow users to record farm-waste information.
	FR12 – Resource Summary	The system shall provide basic summaries of selected farm-resource information.
Persistency	FR13 – Persistent Storage	The system shall store relevant FarmBook information so that it remains available across application sessions.
	FR14 – Retrieve and Update Records	The system shall allow users to retrieve and update previously stored farm information.